4. Dynamics of a particle moving in a straight line or plane

What students need to learn:

The concept of a force. Newton's laws of motion.	Simple problems involving constant acceleration in scalar form or as a vector of the form $a\mathbf{i} + b\mathbf{j}$.
Simple applications including the motion of two connected particles.	Problems may include (i) the motion of two connected particles moving in a straight line or under gravity when the forces on each particle are constant; problems involving smooth fixed pulleys and/or pegs may be set; (ii) motion under a force which changes from one fixed value to another, e.g. a particle hitting the ground; (iii) motion directly up or down a smooth or rough inclined plane.
Momentum and impulse. The impulse-momentum principle. The principle of conservation of momentum applied to two particles colliding directly.	Knowledge of Newton's law of restitution is not required. Problems will be confined to those of a one-dimensional nature.
Coefficient of friction.	An understanding of $\mathrm{F} = \mu R$ when a particle is moving.